Natural Language Processing

Assignment- 9

TYPE OF QUESTION: MCQ

Number of questions: 7 **Total mark: 10 [4*1 + 3*2] (Q5, Q6, Q7 carries two marks each)**

Question 1: Which of the following statements is false?

1. The multinomial distribution models counts over discrete categories.
2. In Latent Dirichlet Allocation (LDA), each topic is represented as a distribution over words.
3. A Dirichlet prior with small α values encourages sparse topic proportions in documents.
4. Increasing the number of topics in LDA guarantees better generalization performance.

Answer: 4

Solution: : Refer to Lecture 43 of Week 9

Question 2: To make each topic focus on a few specific words rather than all words in the vocabulary, which hyperparameter should you adjust?

1. Number of topics K
2. α (alpha)
3. β (beta)
4. Number of documents D

Answer: 3

Solution: β controls topic–word sparsity. Smaller β → topics dominated by fewer words.

Question 3: An LDA model currently has α = 0.7 and β = 0.05. To encourage each document to focus on fewer topics but make topics more general across words, you should:

1. Decrease α, increase β
2. Increase α, decrease β
3. Increase both α and β
4. Decrease both α and β

Answer: 1
Solution:

α: Decreasing → fewer topics per document (sparser document-topic distribution)
β: Increasing →topics cover more words (denser topic-word distribution)

Question 4: Regarding Gibbs sampling in topic modeling, which of the following statements is correct?

1. It cannot estimate the posterior distribution over latent variables.
2. It is a Markov Chain Monte Carlo method.
3. All variables are sampled independently in parallel.
4. Sampling is usually stopped before convergence to the target distribution.

Answer: 2

Solution:
In gibbs sampling, we do sequential sampling until the sampled values approximate the target distribution. This also can directly estimate the posterior distribution over z

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For question 5 , 6 and 7 use the following information.
Suppose you are using Gibbs sampling to estimate the distributions, θ and β for topic models. The underlying corpus has 3 documents and 5 words, **{machine, learning, language, nature, vision}** and the number of topics is 2. At certain point, the structure of the documents looks like the following

$$\beta_i^{(j)} = \frac{C_{ij}^{WT}+\eta}{\sum_{k=1}^{W} C_{kj}^{WT}+W\eta} \qquad \theta_j^{(d)} = \frac{C_{dj}^{DT}+\alpha}{\sum_{k=1}^{T} C_{dk}^{DT}+T\alpha}$$

Doc1: nature(1) language(1) vision(1) language(1) nature(1) nature(1) language(1) vision(1)
Doc2: nature(1) language(1) language(2) machine(2) vision(1) learning(2) language(1) nature(1)
Doc3: machine(2) language(2) learning(2) language(2) machine(2) machine(2) learning(2) language(2)

(number) –number inside the brackets denote the topic no. 1 and 2 denote whether the word is currently assigned to topics t1 and t2 respectively. η = 0.3 and α = 0.3

For question 5,6,7 calculate the value upto 4 decimal points and choose your answer

Question 5 : Using the above structure the estimated value of β(2)nature at this point is

1. 0.0240
2. 0.02459
3. 0.0260
4. 0.0234

Answer: 1

Solution:

	t1	t2
machine	0	4
nature	5	0

language	5	4
vision	3	0
learning	0	3

$\beta(2)$nature = (0+0.3)/(11+5*0.3) = 0.3/12.5 = 0.024

Question 6 : Using the above structure the estimated value of θ_{t1}^{doc2}

1. 0.6562
2. 0.6162
3. 0.6385
4. 0.50000

Answer: 2

Solution:

	t1	t2
doc1	8	0
doc2	5	3
doc3	0	8

θ_{t1}^{doc2} = (5+0.3)/(8+2*0.3) = 5.3/ 8.6 = 0.6162

Question 7 : Using the above structure the estimated value of θ_{t2}^{doc2}

1. 0.6562
2. 0.3975
3. 0.3837
4. 0.3707

Answer: 3

Solution:
Use the same formulae mentioned in Question 9 solution